Mining Software Repositories

Chapter 7: Methods

Prof. Dr. Steffen Herbold

Contents

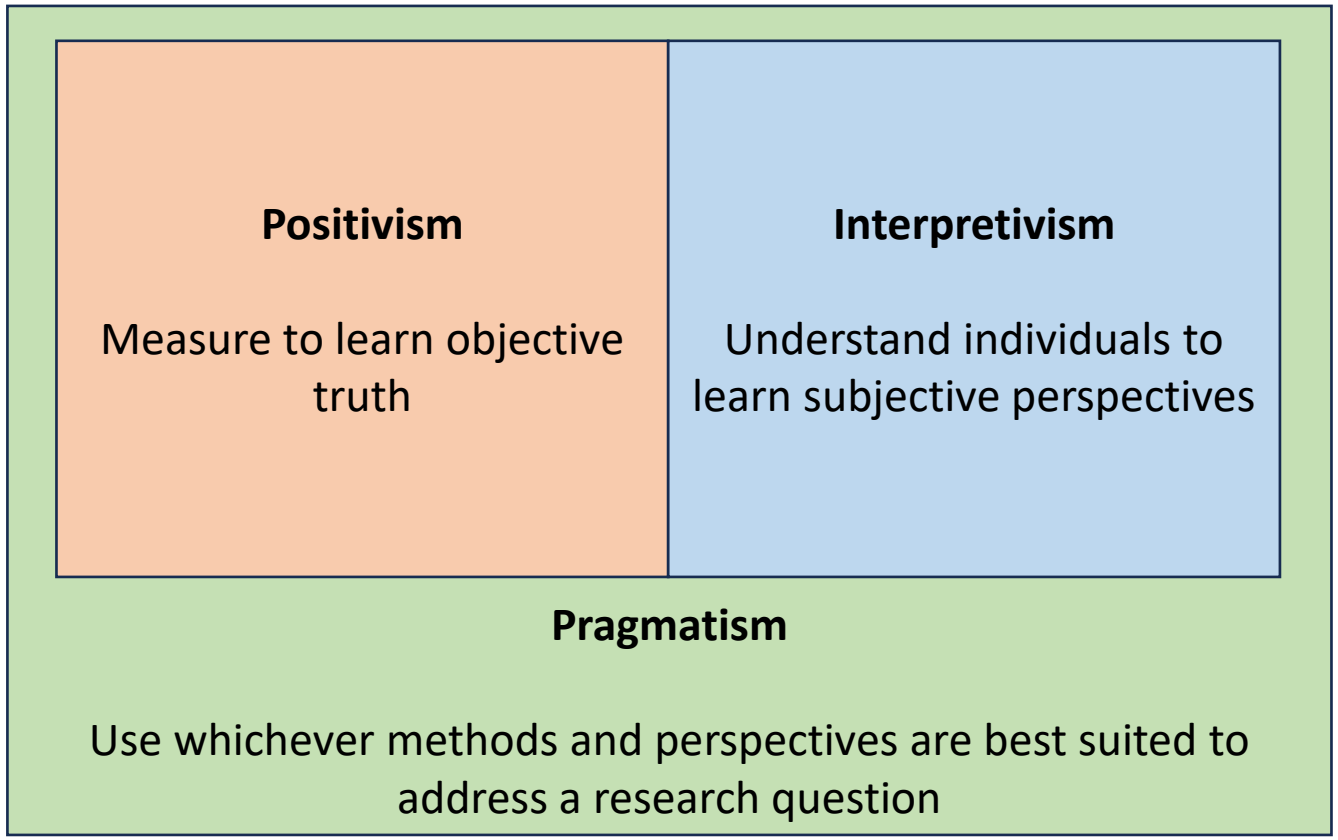
- Research paradigms and research questions
- Sampling
- Basic statistics
- Multiple experiments and paired data
- Multiple variables
- Grounded theory

Research paradigms

- Research methods follow principal paradigms that shape how we select methods and interpret data
- Positivism
  - Reality is singular and can be objectively observed and quantified
  - Knowledge is generated through the empirical observation of evidence
  - Research should be value-free and only based on the results
- Interpretivism
  - Assumes that reality can only be partially known and is – to some degree – constructed in the minds of people
  - Aims to understand subjective meanings that individual attach to their experiences
  - Research should aim to gain a deep understanding of individual perspectives
- Pragmatism
  - Use methods best suited for problem
  - Possibly also mix methods typically associated with positivism or interpretivism

- We do not consider other paradigms (e.g., constructivism or critical realism) here

Relationship between research paradigms



While positivism and interpretivism are mutually exclusive, pragmatism is flexible

MSR and research paradigms

- MSR research is typically positivistic
  - We collect data to measure properties of software development
  - We analyze this data with the intend to come to some objective conclusions

  → Typically quantitative research methods, e.g.,
    - measuring the survival rate of developers within a project to characterize turnover
    - measuring factors of developer involvement to characterize survival rate
- MSR research can also be interpretivist
  - We collect data about human behavior
  - We interpret the data in its context to understand this behavior

  → Typically qualitative research methods, e.g.,
    - understanding how developers react to criticism in code reviews by analyzing review comments
    - characterizing contributions of developers over time to understand their focus and determine reasons why people left a project
- Pragmatic MSR research allows for the combination
  - E.g., to create a quantitative model of the survival rate and understand individual reasons for leaving projects

Formulating research questions

- The way we formulate our research questions are fundamentally tied to our research paradigms

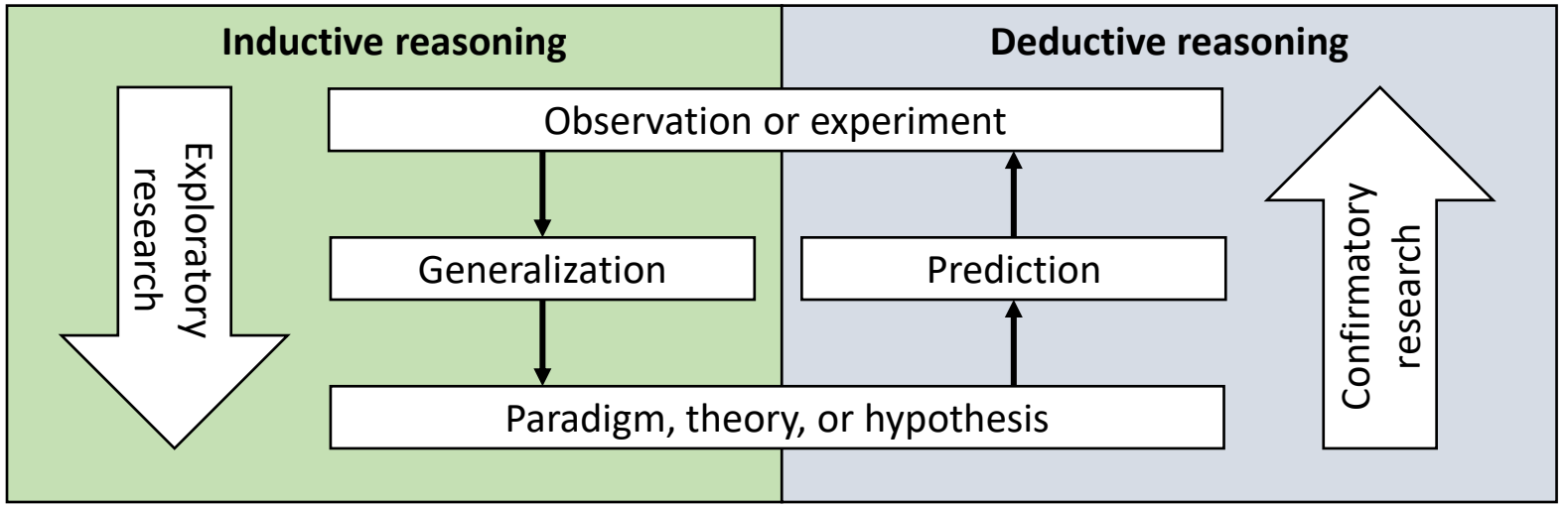
- Positivist research frames research questions as hypothesis or variables to measure
  - The goal is to establish causal relationships or correlations between observed factors

- Interpretivist research typically has a rather open framing with the indent to rather explore meanings, experiences, and perceptions
  - The goal is to understand human behavior in a given context

- Pragmatism allows for both and their combination
  - E.g., trying to quantify behavioral patterns and establish causal relations and at the same time explore the human perspective on this relationship

Hypothesis and reasoning

- A hypothesis is a testable statement that is derived from observed data or an existing theory
- Confirmatory research tries to determine if a hypothesis holds
  - Deductive reasoning, i.e., make predictions based on a hypothesis and see if these predictions holds
  - Typically positivistic, e.g., through the collection of data and testing if this data agrees with the hypothesis
- Exploratory research tries to understand a phenomon with the indent to derive new hypotheses
  - Inductive reasoning, i.e., make inferences from data and try to find generalizations that lead to hypothesis
  - Can be interpretivist or positivistic
    - Depends on the philosophical distinction if the resulting hypothesis should be generally true or rather about individual perspectives



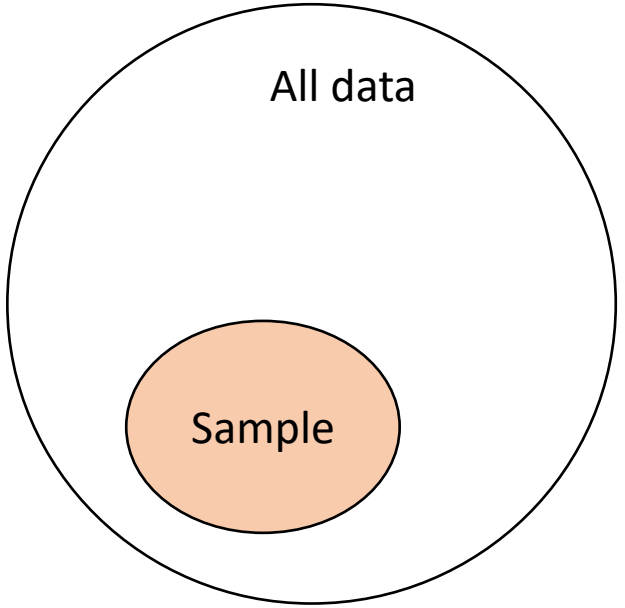
Falsifiability

- A key property of scientific theories (and, as a consequence, hypotheses derived from theories) is whether they are falsifiable
  - In this sense, falsifiable means that we can determine if observations are consistent with the theory
  - Consistent means they agree with the predictions of a theory
  - Inconsistent means they cannot be explained by the theory
- A common problem when defining hypothesis is that they are trivially falsifiable
  - E.g.: „Nesting depth of source code is unrelated to readability of source code“
  - For this hypothesis, a single instance of spaghetti code with many loops and conditions would be sufficient to falsify this
  - However, we would not really have learned anything
- Defining hypothesis in a manner that they are both falsifiable and such that this falsification/non-falsification through observations is a critical tasks of the researcher!
  - Better way to frame the above: „Nesting depth of source code is unrelated to the readability of source code by experts if it is less than or equal to two.“
    - This would study if a shallow nesting levels hinder the understanding of code by experts
  - Alternatively, one could use an exploratory approach first to quantify the relationship between nesting and readability and then formulate a detailed hypothesis based on the results for a subsequent confirmatory study

Sampling

Role of sampling

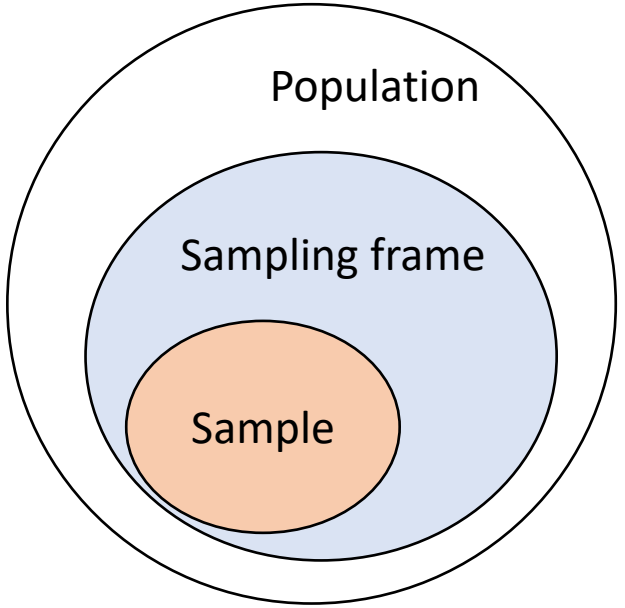
- We usually cannot consider all possible data related to our research in our studies
  - All software projects in existence
  - All open-source projects
  - All software developers
  - All Python programmers
  - …
- Instead, we study subsets of this data
  - Popular projects
  - Projects with a specific size
  - Randomly selected people
- This selection is called sampling



Baltes, S., & Ralph, P. (2022). Sampling in software engineering research: A critical review and guidelines. *Empirical Software Engineering*, *27*(4), 94.

Population, sampling frame, and sample

- When we want to sample, we need to understand what open-source are interested in and what we are sampling from
  - The population are all items that are of interest
    - What these are depends on the research question
  - The sampling frame are the items of interest from which we create a sample
  - The sample are the items of falsifiable interest from the sampling frame that we select through the sampling
  - $\text{Sample} \subseteq \text{Sampling frame} \subseteq \text{Population}$
- Sampling methods define how we pick the sample from the sampling frame
- Example:
  - RQ: What is the male to female ratio within open-source development?
  - Population: All open-source developers
  - Sampling frame: open-source developers on GitHub
  - Sample: 10% the developers of on GitHub selected using simple random sampling



Types of sampling

- In principle, three different approaches for sampling

- Non-probabilistic sampling
  - We select our sample based on fixed criteria such that we have all data that meets these criteria

- Probabilistic sampling
  - We use a randomized method to select a sample

- Multi-stage sampling
  - We use multiple steps for sampling, typically combining non-probabilistic and probabilistic methods to ensure our sample has certain properties, while still looking at a subsample of data

Non-probabilistic sampling (1)

- Convenience sampling
  - Select samples based on availability and/or expedience
    - E.g., projects for which you already have data, human subjects you know or have easy access to
  - Simplest sampling method that usually does not ensure any sample properties
- Purposive sampling
  - Select samples based on a defined strategy
  - The strategy is defined based on the research objectives
  - Criteria are used to operationalize purposive sampling. The criteria cover different aspects, depending on research
    - Popularity (e.g., stars on GitHub)
    - Activity patterns (e.g., number of commits, number of contributors, number of issues)
    - Technology (e.g., programming language, build tools, static analysis methods, software tests)
    - Development process (e.g., use of code reviews, following best practices for linking data)
  - These criteria should not be based on convenience!

Non-probabilistic sampling (2)

- Referral-chain sampling / snowball sampling
  - Requires a starting point, e.g., some samples already available or other information
  - New data points are determined based on their relation to this starting point
  - Example:
    - Start with core machine learning libraries (e.g., scikit-learn, TensorFlow, PyTorch)
    - Then select that are imported by these libraries
      - This would be a sample representing what these libraries depend on

- Respondent-driven sampling
  - Variant of referral-chain sampling for sampling and recruitment of human subjects
  - Start with a small but diverse set of subjects
  - Each subjects is asked to recruit more subjects (e.g., three more people)
    - The idea is, that these recruited people still represent the diversity from the initial sample
  - Repeat until enough subjects are sampled and the results are stable
    - Requires strong mathematical controls that can account for potential sampling bias

Probabilistic sampling

- Simple random sampling
  - Each subject as an equal chance of being sampled, i.e., the chance of inclusion follows a uniform distribution

- Systematic random sampling
  - First order the data points in the sampling frame by some attribute
    - E.g., by popularity, size, age, …
  - Define an interval size, denoted here as $step_size$
  - Randomly select a starting point $start \in [1, step_size]$ with a uniform distribution
  - Select the samples $start, start + step_size, start + 2 \cdot step_size$, …
  - Ensures that subjects are sampled over the whole range of the ordered attribute

Multi-stage sampling (1)

- Stratified and quota sampling
  - Simple random sampling as the drawback that we cannot control what the sample looks
  - While this is intentional, this also might mean we loose important information
    - E.g., when we want to study differences between female and male developers, a simple random sample would likely have (a lot) more male developers than female developers
  - With stratified sampling, we first split our data into subsets based on some property
    - This is the first stage of the sampling
    - E.g., gender, race, bugginess, amount of testing, …
  - Afterwards, we sample independently from the subsets
    - This is the second stage of the sampling
      - If the second stage is probabilistic, this is called stratified sampling
      - If the second stage is non-probabilistic, this is called quota sampling
  - Allows, e.g., to have the same amount of male and female developers in a sample, even if the ratio is not equal in the complete data set

Multi-stage sampling (2)

- Cluster sampling
  - With stratified sampling, we already consider subgroups within the sampling frame
  - Sometimes, it is not feasible to study all these subgroups
    - For example, not papers on MSR from all highly ranked conferences (because there are too many), but only from a subset of the conferences
  - In this case, we can apply cluster sampling
    - First divide the sample into subgroups and sample a subset of these subgroups as clusters to study
    - Then draw samples from the selected clusters
  - If the selection of the clusters and samples are both randomized, this is also called cluster random sampling

- Stratified sampling vs. cluster sampling
  - Stratified sampling uses all subsets, while cluster sampling uses only a subset

Other sampling methods

- Whole frame sampling
  - Whole frame sampling simply uses all data from the sampling frame, i.e., there is no selection of a subset of data

- Panel sampling
  - Panel sampling re-uses a sample multiple times
  - This is often used to study longitudinal effects by observing changes to study subjects
    - E.g., studying the same sample of projects at different points in time

Representativeness

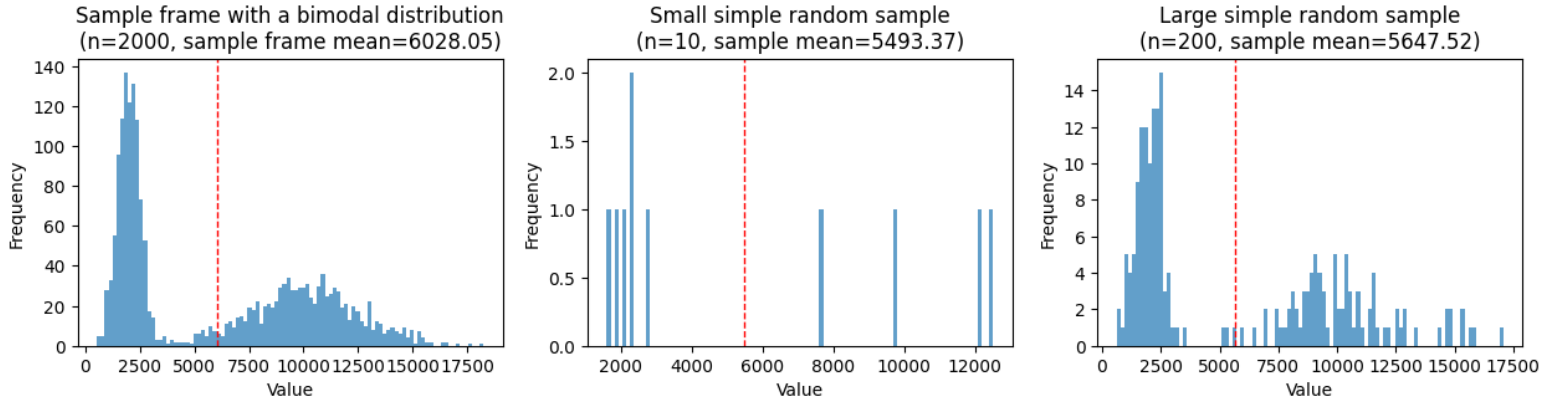
- Definition: Representativeness of a sample
  - The representativeness is the degree to which the properties of interest of a sample resemble these properties within the population.

- Representativeness is important for positivistic research methods
  - Recall: positivism assumes we can objectively observe and quantify phenomena and thereby generate knowledge though empirical observation of evidence
  - Representativeness is important to understand if what we observe in a sample can be generalized to the sampling frame and the population

- Interpretivism is not interested in such generalizations and rather aims to understand data within its subjective context
  - Thus, interpretivism rather accepts that samples may be inherently biased and tries to account for that when interpreting data

Randomness vs. representativeness

- Randomness is often conflated with representativeness
  - „We selected a random sample, therefore it is representative“, „Since the data was not selected randomly, we doubt whether we can generalize“
- While there is a relation, it is not automatic, as the following example shows



- Assume this is data for size of projects within the sample frame
  - The sample frame is bimodal, i.e., we have projects that are small (mean=2000) and large (mean=10000)
  - The small sample is representative for the overall mean, but not for the small and large projects
  - The large sample is representative for the mean and for both small and large projects

Aspects of representativeness

- Representative is dimension-specific
  - A sample can be representative for some properties and not representative for other properties
- Probability sampling does not guarantee representativeness
  - A sample can be unlikely and not follow expected distributions, or an important criterion was not considered during sampling
- Non-probabilistic samples can still be representative
  - A target selection of suitable samples is often helpful, especially for small sample sizes
- Representativeness does not only depend on the type of sampling, but also the size of the sample
  - Problems with probabilistic sampling get smaller with large sample sizes, because it is less likely to have unlikely samples and it becomes more likely that criteria not actively considered during sampling are represented, nevertheless

Arguments for and against representativeness

	Argument	Possible problems
Random selection	Each item in sampling frame as an equal probability of inclusion	Assumes large sample size, good sampling frame, and no bad lack
Size	Sample size is so large that no subpopulations are missed	Can still have biases towards or against subpopulations
Breadth	Sample captures a large variance on important parameters	Point estimates (e.g., mean values) may be unreliable, requires large size to be representative for all parts of covered variance
Parameter matching	Sample and sample frame of similar distribution	Assumes all important properties are known, matching restricted to sample frame

No “one-fits-all“ approach.
Needs to be considered and judged in the context of a concrete study!

Foundations of statistics

Types of statistics

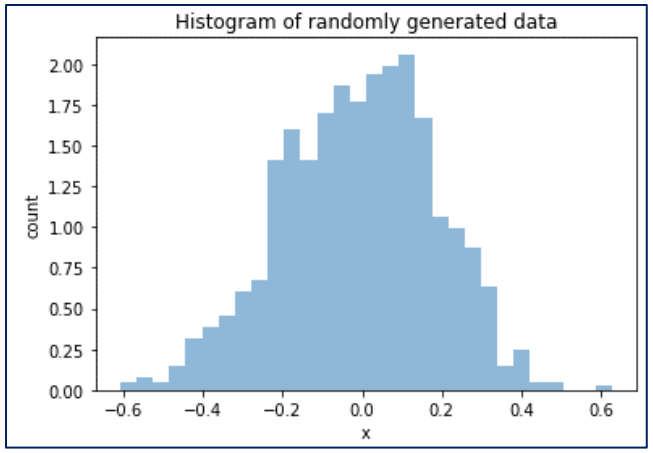
- A key aspect of scientific research is to test if a data is consistent with a hypothesis
- Statistics provide the means to operationalize such tests
  - A statistical model gets data as input and then computes properties about the data
- Descriptive statistics to understand properties of the data
  - E.g., mean, median, standard deviation, skewness, …
  - Can be used to report properties of results or experiments
  - Not suitable for generalization as the properties are only about the observed data
- Inferential statistics tries to understand properties of the probability distribution that generated the data
  - Fitting probability distributions to data or given estimates for parameters of the distribution
  - Suitable for generalization to all data the follows the same underlying distribution
    - Which data this is follows from the representativeness of a sample

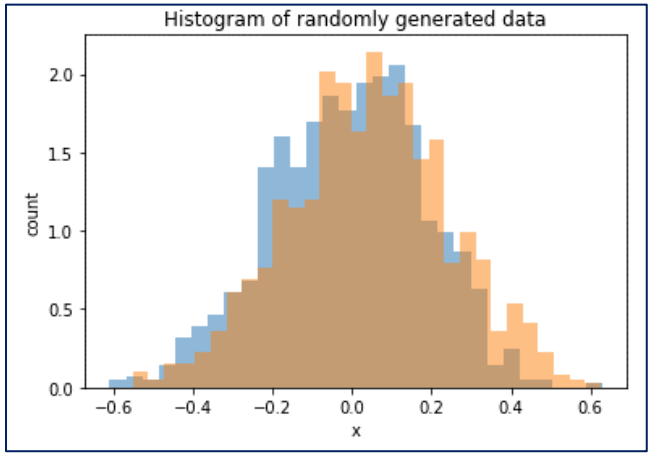
Inferential methods

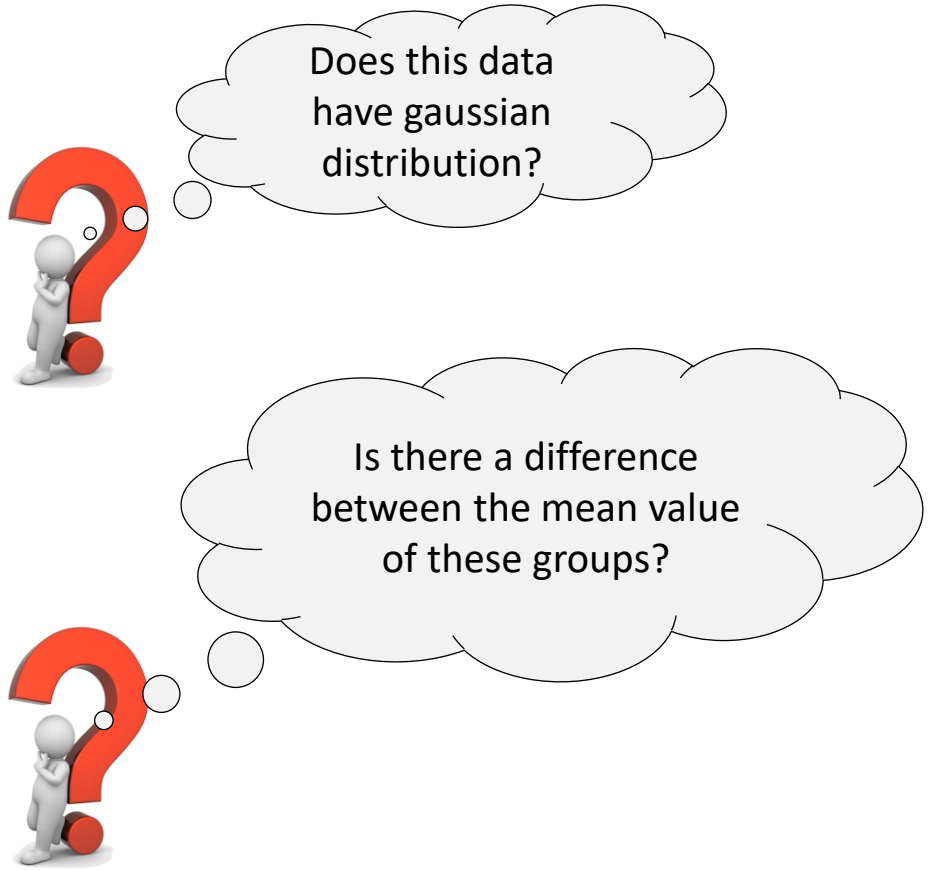
- We consider the following concepts from frequentists statistics
  1) Hypothesis testing
     - Testing if data is consistent with a hypothesis about this data
     - This helps us to understand if observations are significant or likely random
     - Helps to avoid wrong conclusions
  2) Effect sizes
     - Measuring how large differences are between populations to understand the strength of an effect
  3) Confidence intervals
     - Quantifying how certain point estimates (e.g. mean values) are to quantify the uncertainty of results

- Things we do not consider:
  - Statistical power, standard errors, Bayesian methods
  - Linear models are also statistics and considered separately

Reasons for hypothesis testing







Null and alternative hypothesis

- Hypothesis testing evaluates assumptions about data
  - Assumption == Hypothesis

- Null Hypothesis (H_0)
  - Assumption of the test holds and is failed to be rejected at some level of significance.
- Alternative Hypothesis (H_1, H_a)
  - Assumption of the test does not hold and is rejected at some level of significance.

- Most important questions:
  - What is the assumption of a test?
  - What does „rejected at some level of significance“ mean?

P-values

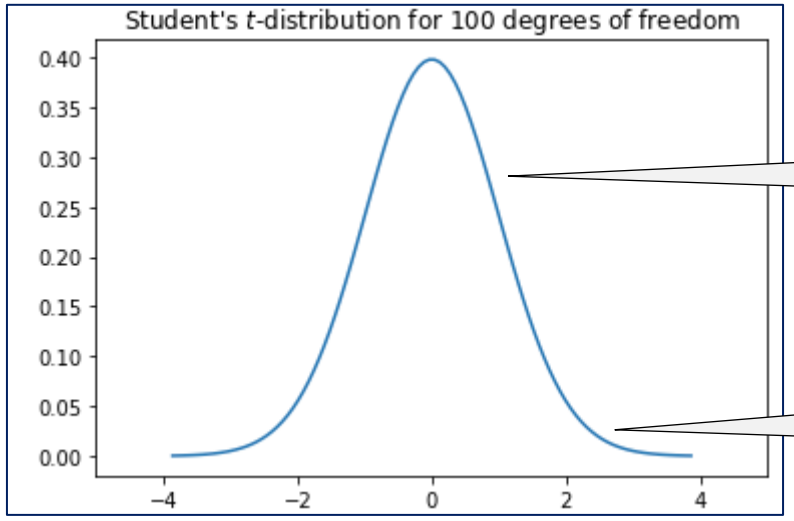
- The probability of the observed data or more extreme data, given the null hypothesis is true.
  - Given the hypothesis, how likely is the data?
  - Not the same as given my data, how likely is the hypothesis!
    - Never use p-values as scores!

- Calculated using the probability density function of a test statistic
  - Given the hypothesis, how likely is this statistical value about the data?

Student‘s t-distribution

- Test statistic used for estimating mean values for normally distributed data

- Probability density function for the location of the deviation of a sample mean value from the real mean value



Roughly bell-shaped

Longer tails than normal distribution

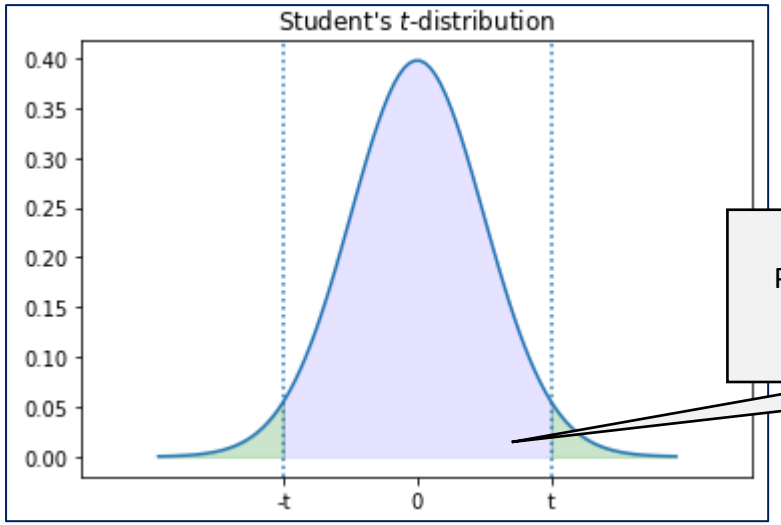
Example: Welch‘s t-test

Also requires calculation of the degrees of freedom, which we omit here



- Null Hypothesis
  - The means of two independent, normally distributed populations X_1 and X_2 are equal.

- t statistic for Welch’s test:
  - $t = \frac{mean(X_1) - mean(X_2)}{\sqrt{\frac{sd(X_1)^2}{N_1} + \frac{sd(X_2)^2}{N_2}}}$



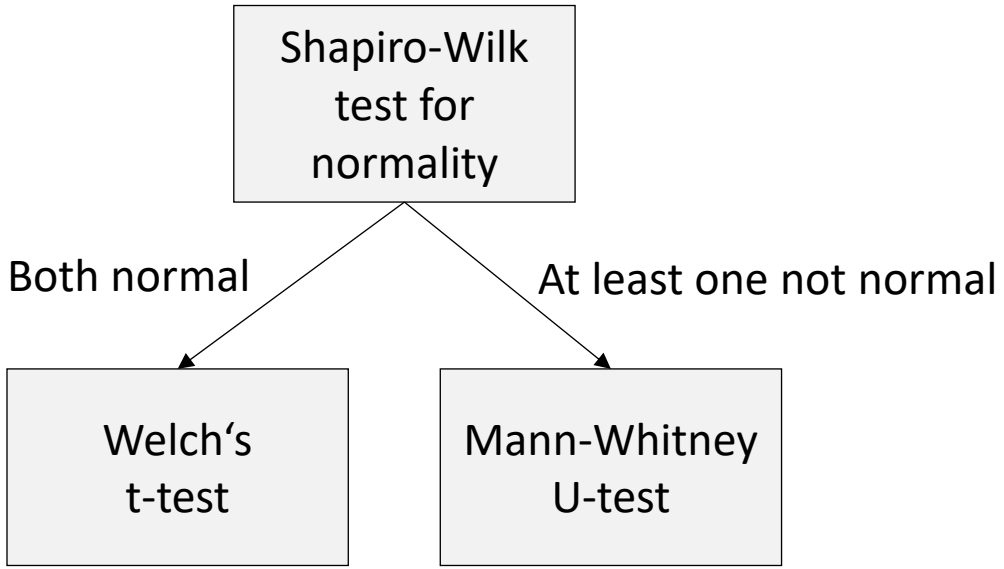
Probability of the observed or more extreme data assuming the null hypothesis is true

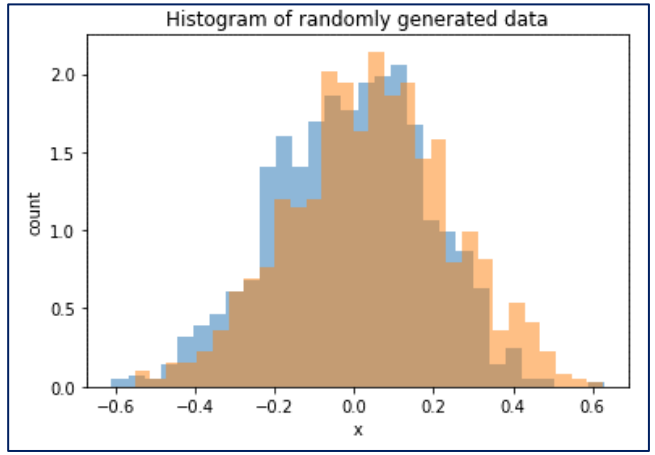
Null hypotheses of important tests

- Welch‘s t-test
  - The means of two independent, normally distributed populations are equal
- Shapiro-Wilk test
  - A population of 3 to 5000 independent samples is normally distributed
  - Anderson-Darling test for larger sample sizes
- Kolmogorov-Smirnoff test
  - Two independent populations have the same probability distribution
- Mann-Whitney-U Test / Wilcoxon-Ranksum-test
  - The values from one population dominate the values of another independent population (more or less difference of means/medians)
- Levene‘s test
  - The variances of a group of populations are equal
- ANOVA
  - The mean values of a group of independent populations are equal
  - Only tells us that there at least one population is different from the others. This is evaluated with a post-hoc test (e.g., Tukey’s HSD).

Combinations of tests

- Example:
  - Are the mean values different?

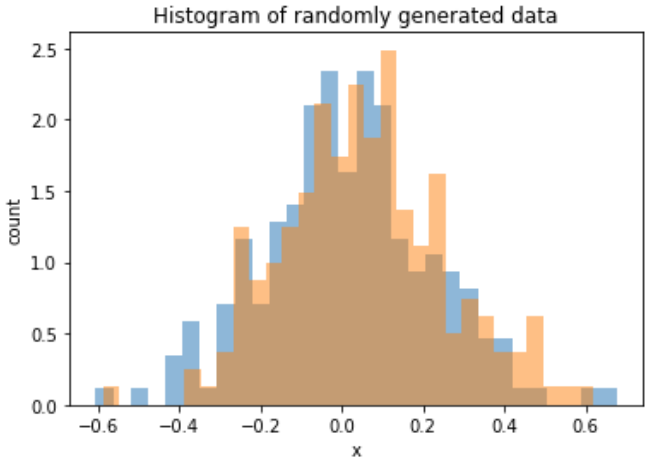




Significance and confidence

- Significance level α
  - Probability of rejecting the null hypothesis, given that it is true
  - Depends on domain
  - Common values:
    - 0.05 (de-facto standard)
    - 0.005 (currently proposed newer standard to reduce false positives)
- Confidence level
  - Probability of not rejecting the null hypothesis, given that it is true
  - $1 - \alpha$
- Used to evaluate tests
  - If $p - value > alpha$ fail to reject null hypothesis
  - If $p - value \leq alpha$ reject the null hypothesis → significant result

Example for running tests




```
p-value of Shapiro-Wilk test for "blue" data:   0.9292
The test found that the data sample was normal, failing to reject the null hypothesis at significance level alpha=0.005

p-value of Shapiro-Wilk test for "orange" data: 0.3986
The test found that the data sample was normal, failing to reject the null hypothesis at significance level alpha=0.005

Both populations normal. Using Welch's t-test.

p-value of Welch's t-tests: 0.048920
The test found that the population means are equal, failing to reject the null hypothesis at significance level alpha=0.005
```

"fail to reject" instead of "accept"

Problems with hypothesis testing

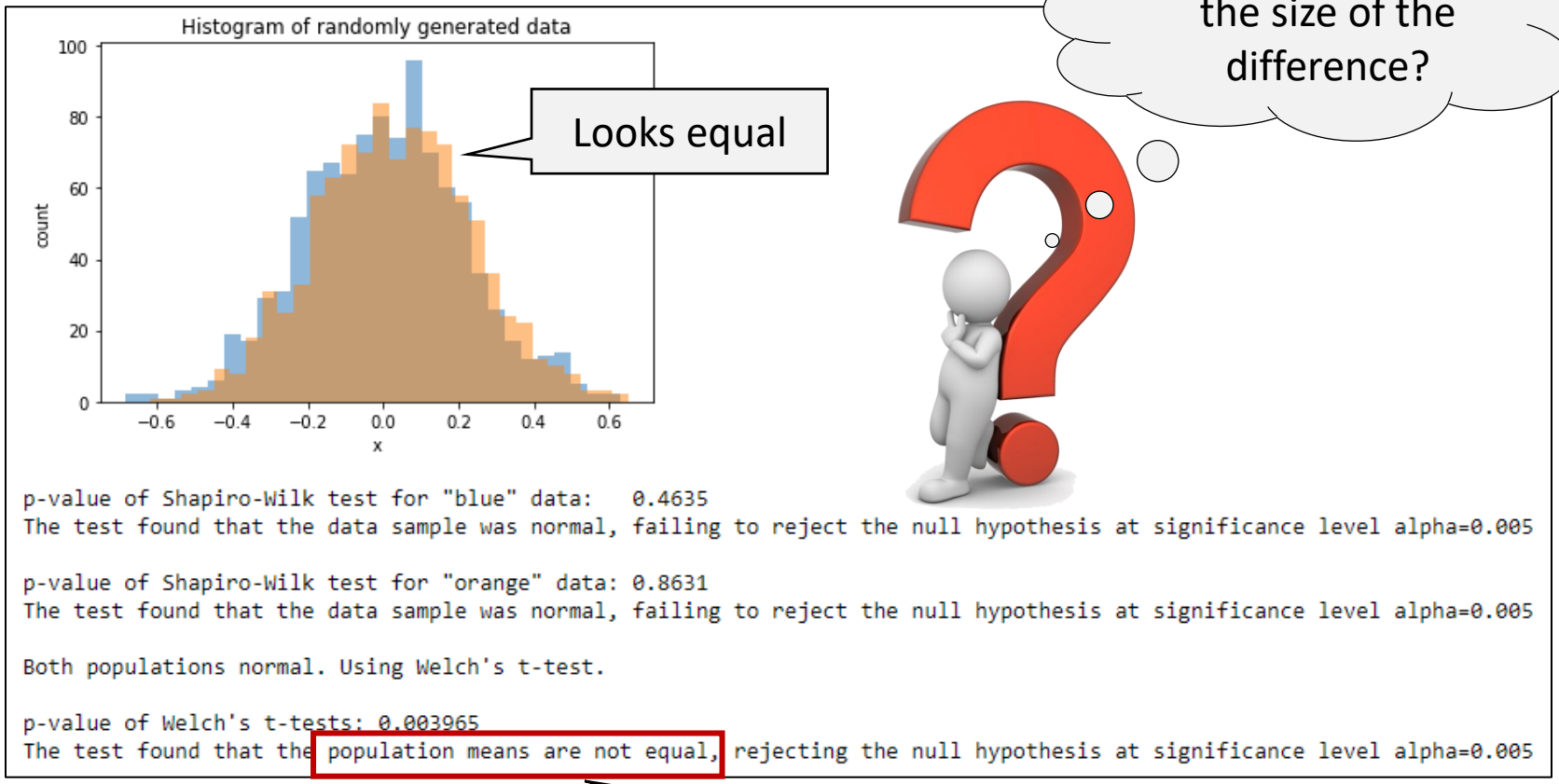
- No binary results!
  - Results are probabilistic, i.e., you can reject/fail to reject the null hypothesis with a certain probability, but you cannot make binary claims.

- Using p-values for scoring
  - p-values describe the likelihood of the data, given the hypothesis
  - Not the same as likelihood of hypothesis, given data!

  → Scoring makes no sense!

- p-hacking
  - Re-running tests with different data until a desired result is found
  - Often inadvertent, e.g., due to subgroup analysis

Significant ≠ important



Effect size

There are also other measures like Hedges' g^*, Glass' Δ that all use a similar concept



- Measures the distance between central tendency with respect to the variance

- Cohen's d
  - Difference of means relative to the standard deviation
    - $d = \frac{mean(X_1) - mean(X_2)}{s}$
    - $d = 1$ means that the difference of means is "one standard deviation"
  - s is the pooled standard deviation
    - $s = \sqrt{\frac{(N_1-1)\cdot sd(X_1)^2 + (N_2-1)\cdot sd(X_2)^2}{N_1+N_2-2}}$

Square root of the weighted mean of the variances

```
Effect size (Cohen's d): -0.129 - small effect
```

How do we know this is small?



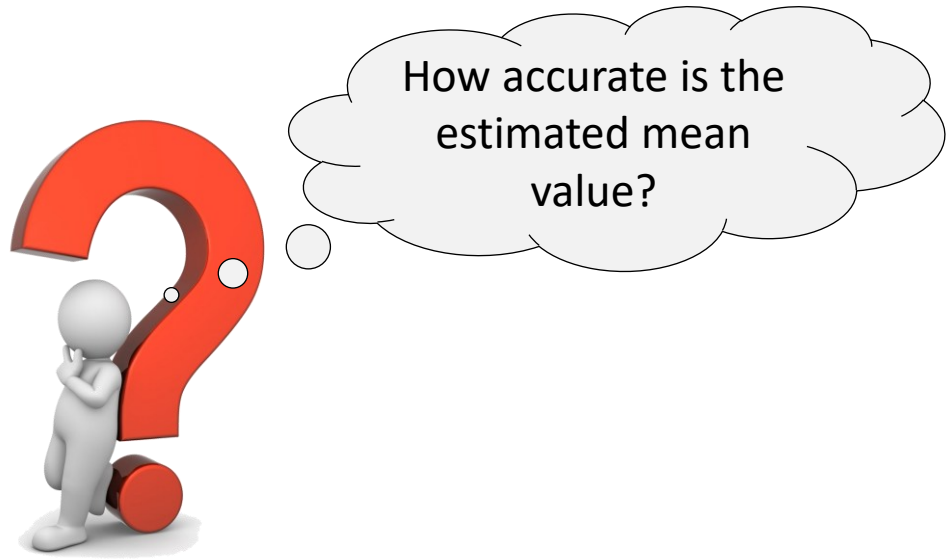
Interpretation of effect sizes

- According to Cohen and Sawilowsky
  - Designed for social sciences
  - Should be used with care, but is broadly used in many domains

Cohen‘s d	Effect size
$\|d\| < 0.01$	Negligible
$\|d\| < 0.2$	Very Small
$\|d\| < 0.5$	Small
$\|d\| < 0.8$	Medium
$\|d\| < 1.2$	Large
$\|d\| < 2.0$	Very Large
$\|d\| \geq 2.0$	Huge

How accurate are estimations?

- Example:
  - You have twenty different data sources.
  - You train on five of them and test on the other fifteen.
  - The mean value of the 15 test values is 0.83, the standard deviation is 0.13.



Confidence intervals

We are assuming that everything is normally distributed. Similar formulas are available for non-normal data



- Definition:
  - A $C\%$ confidence interval θ for some parameter p is an interval that is expected with probability $C\%$ to contain p.
  - C is also called the confidence level

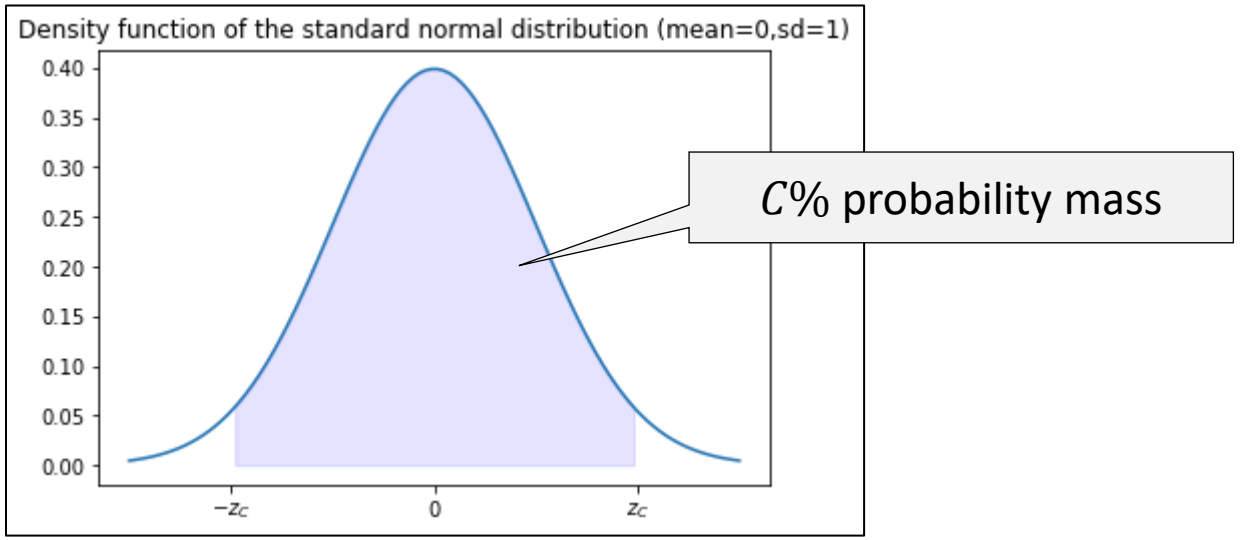
p

- Confidence interval for the mean value of normally distributed data with known standard deviation
  - $\theta = \left[mean(X) - z_C \frac{sd(X)}{\sqrt{n}}, mean(X) + z_C \frac{sd(X)}{\sqrt{n}}\right]$

→ $\pm z_C \frac{sd(X)}{\sqrt{n}}$ uncertainty about the actual mean value

Explanation of z_C

- Choosen such that $[-z_C, z_C]$ is the $C\%$ confidence interval of the standard normal distribution



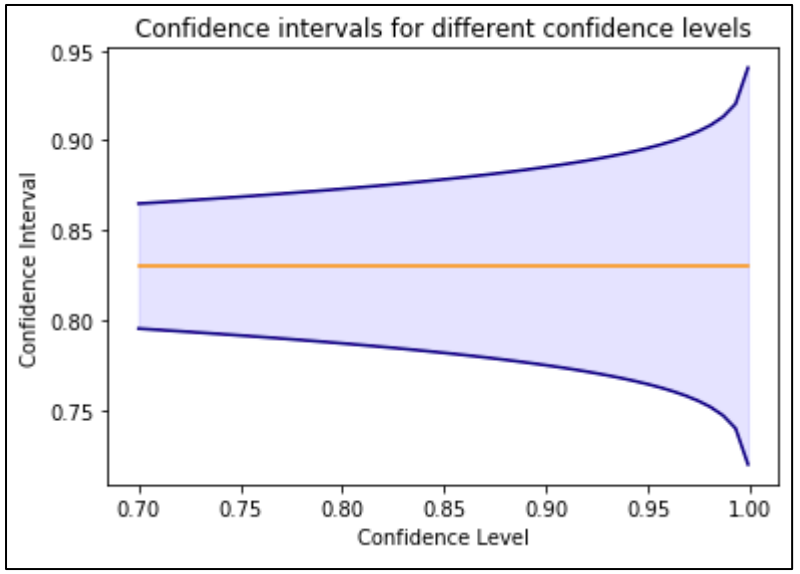
C	z_C
90%	1.645
95%	1.96
99%	2.58
99.5%	2.807
99.9%	3.291

- In practice: look it up in a table

Example for confidence intervals

- Example (repeated):
  - The mean value of the 15 test values is 0.83, the standard deviation is 0.13.

C	z_C	θ
90%	1.645	[0.775, 0.885]
95%	1.96	[0.764, 0.896]
99%	2.58	[0.743, 0.916]
99.5 %	2.807	[0.736, 0.924]
99.9 %	3.291	[0.720, 0.940]



Interpretation of confidence intervals

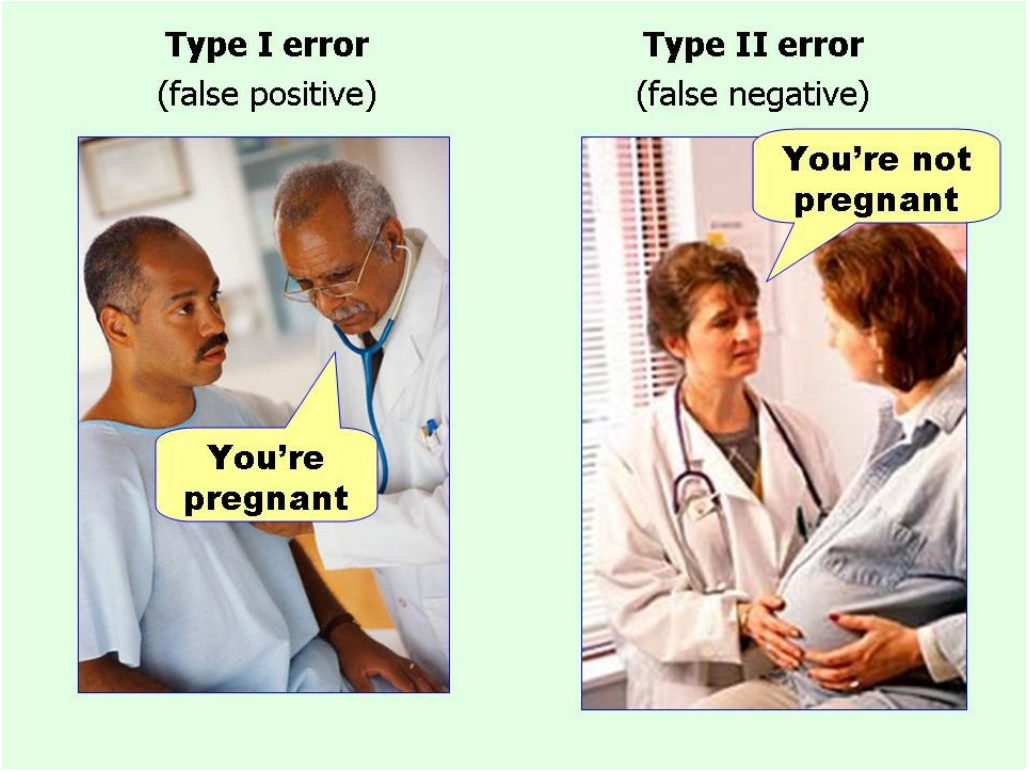
- Correct interpretation of a $C\%$ confidence interval θ
  - $C\%$ chance that results of future replications fall into θ
  - No statistical difference from estimated parameter with $C\%$ confidence
  - There is a $1 - C\%$ probability of the observed data, if the true value for the estimate is outside of θ

- Wrong interpretation
  - The true value lies with $C\%$ probability in θ
  - $C\%$ of the observed data is in θ

Multiple experiments and paired data

Errors of statistical testing

- Type I error (false positive)
  - Wrongly rejecting the null hypothesis
- Type II error (false negative)
  - Wrongly accepting the null hypothesis
- For a single statistical test
  - the probability of a false positive is the significance level α
  - the probability of a false negative is called power and denoted as β
    - The power depends on the sample size and the size of the effect
    - We do not discuss how to calculate the power here, but it should be considered when planning sample sizes of experiments



https://effectsizefaq.com/wp-content/uploads/2010/05/type-i-and-type-ii-errors.jpg

Repeated experiments

- A critical aspect of statistical testing is that the results are not certain

- If we have a significance level of $\alpha = 0.05$ and we run k statistical tests with different data, the probability of having no false positive is $(1 - \alpha)^k$, e.g.,
  - $0.95^{10} \approx 0.60$ for $k = 10$
  - $0.95^{20} \approx 0.36$ for $k = 20$

- We call the expected rate of not making a mistake for many experiments is called the family-wise error rate (FWER)

Correction of p-values

- For the FWER, it is clear that when we have repeated experiments, that the probability of making wrong decisions is not the significance level anymore
  - With many experiments, the opposite is the case: it becomes likely that we sometimes reject the null hypothesis, even if it is true

- To avoid this, we can adjust either the significance level α or the $p - value$

- Bonferroni correction (also known as Bonferroni-Dunn) achieves this for k experiments as follows:
  - Either by adjusting the significance level: $\alpha_{FWER} = \frac{\alpha}{k}$
  - Or by adjusting the outcome of the test: $p - value_{FWER} = p - value \cdot k$
- We then use the adjusted values for making a decision about our hypothesis

- This correction can also be applied to the calculation of confidence intervals

Example for using statistics in MSR

- We want to know if there is an impact of experience on the length and complexity of solutions for a coding task
  - We recruit two groups of developers
    - Group 1 has 20 developers with at least 5 years of experience
    - Group 2 has 20 developers with at most 5 years of experience
  - For both groups, we measure the lines of code (LOC) and the cyclomatic complexity (CC) of the solutions
  - This means we have two populations with sample sizes $n_1 = n_2 = 20$
  - We define two suitable null hypothesis
    - H0-LOC: The mean values for LOC are the same for both groups
    - H1-CC: The mean values for CC are the same for both groups
  - We want to have a Type I error rate of 5%. Since we have two tests, we use $\alpha_{FWER} = \frac{0.05}{2} = 0.025$
  - We run two suitable statistical tests (e.g., t-test if data is normal) to get p-values and accept/reject the null hypothesis based on α_{FWER}
  - We compute the effect sizes for the differences between the mean values in case we find that the differences are significant
  - We compute the confidence intervals for all mean values to report the uncertainty about this point estimate

Independent and paired samples

- Consider an example in which you want to analyze the impact of code review on software quality
  - You estimate the software quality by the number of bugs reported within six months after a release
- Design 1:
  - You collect two samples of projects, one where code review is used, one where it is not used
  - In this case, each project appears only once

Project	Used	#Loc
1	Yes	30
…	…	…
n	Yes	25
n+1	No	32
…	…	…
2n	No	22

- Design 2:
  - You collect data from projects that introduced code review at some time. For each project, you collect the number of bugs once before code review was introduced and once afterwards.
  - In this case, we have each project twice in our data

Project	Used	#Loc
1	Yes	30
…	…	…
n	Yes	25
1	No	32
…	…	…
n	No	22

Statistics with paired data

Project	Used	#Loc
1	Yes	30
…	…	…
n	Yes	25
n+1	No	32
…	…	…
2n	No	22

Leads to two populations where the samples are unrelated with each other
→ **Independent** data!

Project Used	#Loc
1	30
…	…
n	25

Project Not Used	#Loc
n+1	32
…	…
2n	22

Project	Used	#Loc
1	Yes	30
…	…	…
n	Yes	25
1	No	32
…	…	…
n	No	22

Leads to two populations where each sample has a counterpart in the other population
→ **Paired** data
→ Comparisons between populations should be based on comparison of pairs

Project	#Loc Used / Yes	#Loc / Used No
1	30	32
…	…	…
n	25	22

Statistical tests for paired data

- We use paired difference tests for paired data instead of the tests we introduced before

- Example: Paired t-test
  - Null Hypothesis
    - The means of two paired, normally distributed populations $X_1 = \{x_1^1, \dots, x_n^1\}$ and $X_2 = \{x_1^2, \dots, x_n^2\}$ are equal.
  - We first create a new population as the difference between the two original populations X_1 and X_2
    - $\bar{X} = \{x_1^1 - x_1^2, \dots, x_n^1 - x_n^2\}$

Project	#Loc Used / Yes	#Loc / Used No	Difference
1	30	32	2
…	…	…	…
n	25	22	-3

  - We can the use a one-sample t-test to determine statistic for the paired t-test:
    - $t = \frac{mean(\bar{X})}{sd(\bar{X})/\sqrt{n}}$
    - This tests if the difference is significantly different from zero

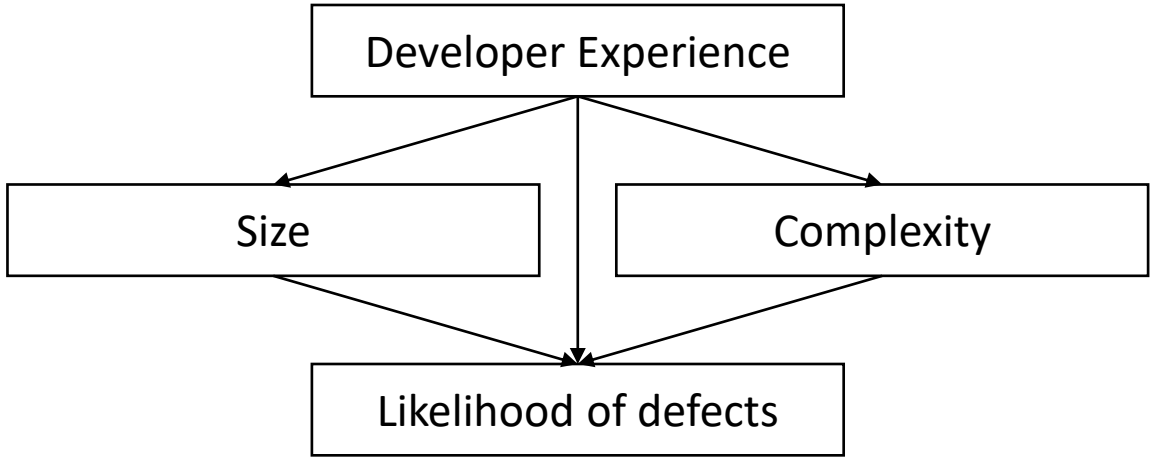
Between-subjects and within-subjects designs

- In our last example, the difference between the two designs was that in one, we had different projects in the populations and in the other, we had multiple measurements for the same projects
  - These types of design are also called between-subjects and within-subjects design
- In a between-subjects design, every sample (i.e., our subjects) is used only once
  - Independence makes design often simpler
  - Subjects have single role, avoids carry-over effects
  - Requires good controls that the different groups of subjects are sufficiently similar
    - Since we have two groups of subjects that are independent, we need to ensure that both are representative for all properties of interest
    - This includes confounding factors, as these could otherwise also explain differences
- In a within-subjects design, every sample is used multiple times under different conditions
  - Naturally produce paired data, i.e., pairing by subject
  - Often more efficient as fewer unique samples are required
  - Enables direct comparison which can help to avoid confounding effects
- More considerations required when human subjects are used, as there may be carry-over effects
  - Not discussed here

Multiple variables

Modeling with multiple variables

- The statistics we covered so far are suitable to analyze a single variable
  - E.g., we can test for the difference between two mean values
- We often want to know about the relationship between multiple variables
  - For example, we want to know how complexity and size together affect the likelihood of defects
- Moreover, we often want to control for possibly confounding effects
  - For example, a relationship between size and complex



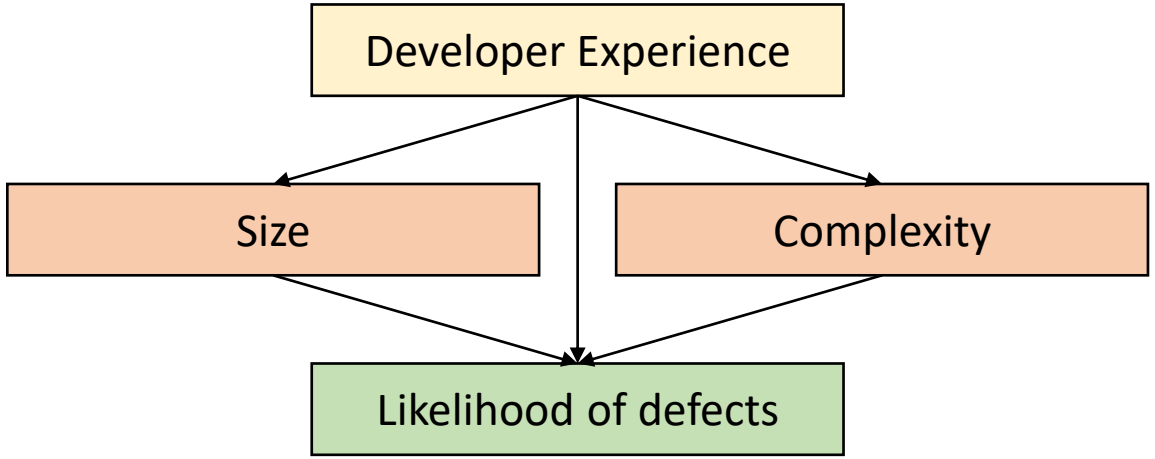
Arrows depict that a variable may influence another variable

Types of variables

Confounder
Independent
Dependent

- In principle, this means we consider three types of variables in experiments

- Dependent variable (denoted as Y)
  - Outcome of interest for the study, also called response variable
  - We want to know how the other variables influence this variable



- Independent variables (denoted as $X_1, \ldots, X_p$)
  - Variables that we expect that influece the dependent variable

- Confounding variables / control variables (denoted as $Z_1, \ldots, Z_q$)
  - Variables that we believe may be related to both dependent and independent variables
  - Used to control that the relationship between dependent/independent variables is really between these variables and not explained by the confounding variable

Linear models

- With a linear model, we model the relationship between the variables through a linear function
  - $Y = \beta_0 + \sum_{i=1}^{p} \beta_i X_i + \sum_{j=1}^{q} \gamma_j Z_j + \epsilon$

- The parameters of this model can be studied to understand the relationships
  - The intercept β_0 can be interpreted as the value of the dependent variable, if all independent and confounding variables are zero
  - The coefficients $\beta_1, \dots, \beta_p$ model the relationship between the independent variables and the dependent variable
  - The coefficients $\gamma_1, \dots, \gamma_q$ model the relationship between the confounding variables and the dependent variable
  - The error term ϵ models the residual error in the prediction
    - $\epsilon = Y - \left(\beta_0 + \sum_{i=1}^{p} \beta_i X_i + \sum_{j=1}^{q} \gamma_j Z_j\right)$
    - Often not written as part of the model but rather computed after fitting the linear model

Example for a linear model

- We have seen linear models before
  - E.g., when we discussed the relationship between gender ratio and community smell in Chapter 5

Factor	All Variables		
	Estimate	S.E	Sig.
(Intercept)	-2.260	0.635	
log(totalCommitters)	-0.049	0.085	
log(totalcommits)	0.027	0.060	
projectAge	-0.025	0.015	.
turnover	10.048	0.455	***
blauGender	0.194	0.804	
tenureMedian	0.044	0.035	
tenureDiversity	0.010	0.025	
stCongruence	-0.308	0.186	.
centrality	-0.038	0.146	
log(teamSize)			
truckFactor	-0.014	0.046	
numberOfWomen	0.038	0.012	**
****p < 0.001 - **p < 0.01 - *p < 0.05 .p < 0.1*			

β_i

γ_j

- Since coefficients are almost never exactly zero, we need to use statistical test to determine if they are significant
  - This means if their difference from zero can be explained by random effects

Significance of coefficients

- To understand if the coefficients are significant, we need to answer two questions
- First, does the model explain the dependent variable or is the result not much better than randomness?
  - Null hypothesis: The coefficients of the independent and confounding variables are equal to zero
  - Alternative hypothesis: At least one coefficient is not equal to zero
  - This can, e.g., be evaluated using an ANOVA test
    - Further assumes the data for all variables is independent and normal
- Second, in case we reject the null hypothesis, we we need to figure out which coefficients are significant
  - These are multiple tests (one for each coefficient) and require correction
  - Dedicated tests that implement this, e.g., Tukey‘s HSD for ANOVA
    - Basically a computation of confidence intervals that corrects for considering all pairs of variables
    - Coefficient is significant, if zero is not in the confidence interval

- In such a two-stage scenario we call the first test the omnibus test (is there any difference?) and the second test the post-hoc test (what are the difference?)

Generalized linear models

- Linear models assume that the relationship between the variables is linear, and that the response variable is continuous
- They further assume independence between the variable, as otherwise the interpretation of the coefficients can be misleading
  - Regularization (e.g., Ridge, Lasso) allows to account for this, but this hinders statistical analysis

- Generalized linear models
  - Introduces a link function g
    - Models $\mathrm{g}(Y) = \beta_0 + \sum_{i=1}^{p} \beta_i X_i + \sum_{j=1}^{q} \gamma_j Z_j + \epsilon$
  - The link function can be interpreted as a transformation of the dependent variable
    - One example for a link function is the logistic function that is used for logistic regression
    - This allows the use of linear models with binary response variables
    - Other link functions can be used to have models for, e.g., count data or skewed data
  - Other assumption (linear relationship with respect to link function, problems with correlations) stay the same

Non-linear models

- There are statistical models that allow us to model non-linear relationships and still test for significance
  - For example, Generalized Additive Models (GAMs) can be seen as an extension of linear models in which we also apply functions to the independent/confounding variables
  - Require a manual modeling of the non-linearities
    - I.e., these must be known and are not “learned“
    - Requires high expertise in both statistical models and the domain for which the model is created

- Machine learning allows the discovery of complex relationships between variables
  - These relationships can be interpretable by humans (e.g., decision trees) or black boxes (e.g., random forests, neural networks)
  - Interpretable models allow us to understand the structure of the relationships
  - Black boxes in general only allows us to understand that there is a relationship
    - Some analysis is still possible, depending on models
    - E.g., feature importance for tree-based models to understand which variables were used or XAI techniques like LIME or SHAP
  - The main drawback of using machine learning models is that it is in general not possible to know if relationships are significant or rather random correlations found by the learning algorithm

  - An example for this approach was used in Chapter 6 for looking at the relationship between costs and performance metrics

Grounded theory

Inductive theory generation

- The research methods we considered so far were based on statistics

- Statistics are a powerful method to analyze hypothesis that you already have
  - This requires some existing theory from which the hypotheses are derived
  - Best suited for confirmatory research
  - Can also be used inductively, but in this case the results are less trustworthy
    - E.g., by running many tests to see if any variables have a relationship without having a theory yet what the relationships might be
    - Results of tests can then be used to create such a theory inductively
  - Statistics assume a positivistic world view, i.e., that we can capture reality through these numbers

- What we are still lacking are methods for interprevistic research that are suitable to generate theories
  - Note: we do not consider methods for interprevistic confirmatory research (e.g., triangulation)

- As example for such a method, we consider grounded theory

Stol, K. J., Ralph, P., & Fitzgerald, B. (2016, May). Grounded theory in software engineering research: a critical review and guidelines. In *Proceedings of the 38th International conference on software engineering* (pp. 120-131).

Principles of grounded theory

- Grounded theory is a set of methods with different variants proposed over the years, e.g.,
  - Classic / Glaserian grounded theory (Glaser and Strauss, 1967)
  - Straussian grounded theory (Strauss and Corbin, 1990)
  - Constructivist grounded theory (Charmaz, 2000)

- There are more variants and for each of these variants there are multiple sources and updates over the years
  - Within this lecture, we focus on Straussian grounded theory as published in 1998
  - Glaserian grounded theory puts a stronger focus on the *emergence* of a theory only from the data without prior knowledge, biases and expectations
  - Constructivist grounded theory is somewhere in the middle, while also allowing for a constructivist stance
  - The variants also have difference between the way data is analyzed and the terminology, including between evolution of variants

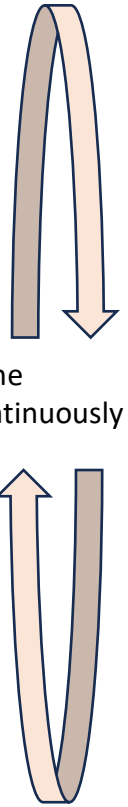
Glaser BG, Strauss A. Discovery of Grounded Theory. Strategies for Qualitative Research. Sociology Press, 1967

Strauss, A., & Corbin, J. M. (1990). *Basics of qualitative research: Grounded theory procedures and techniques.* Sage Publications, Inc.

Strauss, A. and Corbin, J. 1998. Basics of Qualitative Research: Techniques and Procedures for Developing Grounded Theory. Sage, 2nd Ed.

Charmaz, K. 2000. Grounded theory: Objectivist and contructivist methods. In The Handbook of Qualitative Research. Edited by N. K. Denzin and Y. Lincoln. Thousand Oaks, CA: Sage Publications, Inc.

Overview of Straussian grounded theory



Input: (Usually textual) data
In MSR, e.g., issues or code reviews collect from a repository

Open coding

Analyze the data and inductively identify concepts
Concepts are described by code words

Axial coding

Identify relationships between the categories
This is based on the context, conditions, actions/interaction strategies, and consequences

Selective coding

Determine overarching major categories that all other can link to
Should be abstract and suitable to develop hypotheses

Forming theory

Formulate a theory that explains the identified relationships
Theory should be based on the (major) categories

Output: Theory that is developed based on the data
E.g., a theory about how developers interact within code reviews

Memoing

- Throughout the process, researchers keep and update notes called memos
- The memos can be diverse and support the analysis activities
- Examples for memos are
  - Explanations why a certain category was assigned
  - Explanations for the codes that describe the categories
  - Diagrams showing relationships between categories
  - Gaps within the current data, categories, relationships, or theory
- Memos are crucial for all grounded theory activities and form the backbone of the inductive reasoning
  - The memoing should be on a sufficient level of detail such that other researchers can follow all reasoning steps

Development of research questions

- The positivistic research methods we considered earlier assume that there are clear research questions
  - Grounded theory breaks with this assumption

- Within Straussian grounded theory, the research question could be
  - Developed during the coding process
  - Defined beforehand based on a clear objective
  - Derived from existing literature, e.g., as an open questions about a topic or something that should be confirmed
    - This enables grounded theory as a method for confirmatory research

- Research questions for grounded theory are typically more open and generic than those for positivitic research
  - Positivism typically comes with an expectation, at least on the hypothesis level
  - The interpretivistic grounded theory (usually) does not have such expectations but rather lets this be guided by the data

The role of literature

- A key aspect of grounded theory is that the theory should be derived from the data, not from the literature
  - Exposure to literature means that existing knowledge and theory influence the researcher and, consequently, also the developed theory
  - Can lead to confirmation bias

- The original Glaserian variant of grounded theory suggested not to consider literature at the beginning at all
- Later variants, including Straussian grounded theory allow for early considerations of literature, including before the start of a study for the development of research questions
  - Straussian grounded theory allows the use of literature throughout the process to help with a better theoretical understanding, as secondary data source
  - All variants state that literature should be used to enhance the emerging theories

Theoretical sampling and saturation

- As part of grounded theory, researchers can identify the need for more data
  - To get further information about gaps found in the already collected data
  - To get more data for specific categories

- This process of additional data collection based on the needs to develop the theory is called theoretical sampling
  - This is a type of purposive sampling

- This data collection stops once theoretical saturation is reached
  - Achieved when all components of the theory are well supported
  - New data is no longer added information and leading to revisions of the theory

Notes on terminology

- Other variants of grounded theory use different variants of coding, e.g.,
  - theoretical coding (Glaserian, constructivist), which is similar to axial coding
  - initial coding (constructivist), which is similar to open coding
  - focused coding (constructivist), which is similar to selective coding but rather focuses on the most frequent categories

- The usage of the term category can also be different
  - Some source first develop *initial codes*, which are then grouped into *concepts*, which are further refined in to *categories*
  - When this terminology is used, the concepts are typically the result of the open coding and the categories the result of the axial coding

Questions?

